Angles in Polygons

Exam code: 0607

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Angles in polygons

What is a polygon?

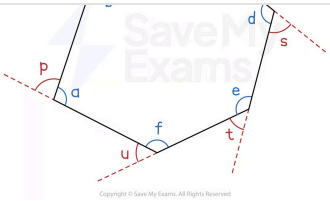
- A **polygon** is a 2D shape with n straight sides
  - A **triangle** is a polygon with **3 sides**
  - A **quadrilateral** is a polygon with **4 sides**
  - A **pentagon** is a polygon with **5 sides**
- In a **regular** polygon **all the sides are the same length** and **all the angles are the same size**
  - A **regular polygon** with **3 sides** is an **equilateral triangle**
  - A **regular polygon** with **4 sides** is a **square**

What are the interior angles and the exterior angles of a polygon?

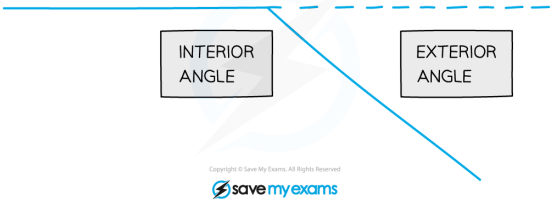
- **Interior angles** are the angles **inside** a polygon at the corners
- The **exterior angle** at a corner is the angle needed to **make a straight line with the interior angles**
  - It is **not** the angle that forms a **full turn** at the corner

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- The **interior** angle and **exterior** angle **add up to 180°** at each corner



What is the sum of the interior angles in a polygon?

- To find the **sum of the interior angles** in a polygon of n sides, use the rule
  - Sum of interior angles = $180° \times (n - 2)$
    - This formula comes from the fact that n-sided polygons can be split into $n - 2$ triangles
- **Remember** the sums for these polygons
  - The interior angles of a **triangle** add up to **180°**
  - The interior angles of a **quadrilateral** add up to **360°**
  - The interior angles of a **pentagon** add up to **540°**

What is the sum of the exterior angles in a polygon?

- The **exterior angles** in **any** polygon always **sum to 360°**

How do I find the size of an interior or exterior angle in a regular polygon?

- To find the **size of an interior angle** in a **regular polygon**:
- **Method 1**
  Find the **sum** of the **interior angles**
  - For a pentagon: $180° \times (5 - 2) = 540°$
  - **Divide** by the **number of sides** (n)
    - For a pentagon: $540° \div 5 = 108°$
- **Method 2**
  Use the **formula** $\frac{180(n-2)}{n}$ which calculates the interior angle directly
- To find the **size of an exterior angle** in a **regular polygon**:
  - either divide 360° by the number of sides (n)
    - For a pentagon: $360° \div 5 = 72°$

- **Subtract** the **exterior** angle from **180°** to find the **interior** angle
- **Subtract** the **interior** angle from **180°** to find the **exterior** angle

Regular Polygon	Number of Sides	Sum of Interior Angles	Size of Interior Angle	Size of Exterior Angle
Equilateral Triangle	3	180°	60°	120°
Square	4	360°	90°	90°
Regular Pentagon	5	540°	108°	72°
Regular Hexagon	6	720°	120°	60°
Regular Octagon	8	1080°	135°	45°
Regular Decagon	10	1440°	144°	36°

How do I find a missing angle in a polygon?

- To find a **missing angle** in a polygon:
  - Use the formula $180° \times (n - 2)$ to work out the **sum** of the interior angles
  - **Subtract** the **other interior angles** in the polygon

How do I find the number of sides in a regular polygon?

- If you are given the interior angle of a regular polygon
  - set the angle equal to $\frac{180(n-2)}{n}$
  - then solve the equation to find n

Examiner Tips and Tricks

Make sure you identify whether you are dealing with a **regular** or **irregular** polygon before you start a question.

Finding the sum of the interior angles using $180 \times (n - 2)$ can often be a good starting point for finding missing angles.

The exterior angle of a regular polygon is 45°.

Write down the name of the polygon.

Answer:

The formula for the exterior angle of a regular polygon is

$$\text{Exterior Angle} = \frac{360°}{n}$$

Substitute the 45 for the exterior angle

$$45° = \frac{360°}{n}$$

Solve by rearranging

$$n = \frac{360}{45}$$

$$n = 8$$

Write down the name of a shape with 8 sides

Regular octagon

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